

It has five Research Centres

Name	Year	Location
Bhabha Atomic Research Centre (BARC)	1954	Trombay, Mumbai
Raja Ramanna Centre for Advanced Technology (RRCAT)	1984	Indore (Madhya Pradesh)
Indira Gandhi Centre for Atomic Research (IGCAR), Kalpakkam	1971	Tamil Nadu
Variable Energy Cyclotron Centre (VECC)	1977	Kolkata
Atomic Minerals Directorate for Exploration and Research (AMD)	1998	Hyderabad

Nuclear Power Plants in India

Power Station	State
Tarapur	Maharashtra (India and USA)
Rawatbhata	Kota, Rajasthan (India and Canada)
Kudankulam (Tirunelveli)	Tamil Nadu (India and Russia)
Kaiga	Karnataka
Kalpakkam	Tamil Nadu
Kakrapar	Gujarat
Narora	Uttar Pradesh

India's Nuclear Test

- On 18th May, 1974 India conducted her first underground nuclear explosion at Pokhran (Rajasthan) in the Thar desert at a depth of 100 metres. The code name used to convey the success of the test to the then Prime Minister, Indira Gandhi, was 'Buddha is Smiling'.
- For the second time, India conducted on 11th May, 1998 three underground nuclear explosions at the same place *i.e.*, Pokhran in the Thar desert of Rajasthan at a depth of 100 metres. The test were code named 'Operation Shakti'.
- The Indian Government has officially declared the 11th May as National Technology Day in India to commemorate the first of the five nuclear tests that were carried out on 11th May, 1998. The day is celebrated by giving awards to various individuals and industries in the field of science and technology.

INDIAN DEFENCE

- The President of India is the Supreme Commander of the Indian Armed Forces (Army, Navy and Air Force).
- Administrative and operational control is with the Defence Ministry of India.

The Ministry of Defence comprises of four departments, which are as follows :

- Department of Defence
- Department of Defence production
- Department of Defence Research and Development (DRDO)
- Department of Ex-Service man welfare

Field Formations of the Indian Army

Division	It usually consists of 15000 combat troops and 8000 support elements. Each division is composed of several Brigades.
Brigade	A Brigade generally consists of around 3000 combat troops with supporting elements. An Infantry Brigade usually has 3 Infantry Battalions along with various Support Arms and Services.
Battalion	A Battalion is commanded by a Colonel and is the Infantry's main fighting unit. It consists of more than 900 combat personnel.
Company	Headed by the major a company comprises 120 soldiers.
Platoon	An intermediate between a company and section a Platoon is headed by a Lieutenant. It has a total strength of about 32 troops.
Section	Smallest military outfit with a strength of 10 Personnel.

Indian Army

HQ	New Delhi
Motto	Service Before Self
Battle Cry	Bharat Mata ki Jai
Founded on	1947
Army Day	15th January
Active Personnel	1.4 million
Reserve Personnel	2.1 million
Colours	Gold, Red and Black
First Army Chief	Robert Mc Gregor Lockhart (1947)

- Indian Army is organised into seven Regional Commands

Army Commands and their Headquarters

Army Commands	Headquarters
Northern Command	Udhampur
Southern Command	Pune
Eastern Command	Kolkata
Western Command	Chandi mandir (Panchkula, Haryana)
Central Command	Lucknow
South-Western Command	Jaipur
Army Training Centre (ARTRAC)	Shimla

- Vairengte is the Indian Army's counter-insurgency school.

Commissioned Ranks of Three Forces

The following are the commissioned ranks in the three services. Each rank is shown opposite its equivalent in the other service.

Army	Navy	Air Force
General	Admiral	Air Chief Marshal
Lieutenant General	Vice-Admiral	Air Marshal
Major General	Rear Admiral	Air Vice-Marshal
Brigadier	Commodore	Air Commodore
Colonel	Captain	Group Captain
Lieutenant Colonel	Commander	Wing Commander
Major	Lieutenant Commander	Squadron Leader
Captain	Lieutenant	Flight Lieutenant
Lieutenant	Sub-Lieutenant	Flying Officer

Indian Navy

HQ	New Delhi
Motto	Sham no varunah
Navy Day	4th December
Personnel	79,023
Founded	1947

Functional Commands and Headquarters

Navy Commands	Headquarter
Eastern Command	Visakhapatnam
Western Command	Mumbai
Southern Command	Kochi

- *Kanwar is the operational Naval Base which is exclusively controlled by Indian Navy. Hindustan shipyard is located at Visakhapatnam. Goa shipyard is in Vasco-da-Gama.*

Indian Airforce

Founded	8th October, 1932
Air Force Day	8th October Motto - Nabha Sparsham Deeptam (Touch the Sky with Glory)
HQ	New Delhi
Personnel	140,139
Colours	Navy blue, sky blue and white

Functional Commands and Headquarters

Training Command (TC)	Bengaluru, Karnataka
Maintenance Command (MC)	Nagpur, Maharashtra

Operational Commands and Headquarters

Central Air Command (CAC)	Allahabad, Uttar Pradesh
Eastern Air Command (EAC)	Shillong, Meghalaya
Southern Air Command (SAC)	Thiruvananthapuram, Kerala
South Western Air Command (SWAC)	Gandhinagar, Gujarat
Western Air Command (WAC)	Subroto park, New Delhi

Territorial Army (TA)

It is a citizens voluntary force, which functions on a part time basis. It was established on 9th October, 1949 and is designed to give the citizens an opportunity to receive military training during their spare time.

Internal Security of India

Indo-Tibetan Border Police (ITBP)	It was established in 1962, after the Chinese attack. It protects India-China border.
National Security Guards (NSG)	Mandate to fight in anti-terrorist operation and in security of VVIPs.
Central Industrial Security Force (CISF)	It was set-up in 1969, its work is to secure industrial complex of government.
Assam Rifles	It was established in 1835 and is the oldest paramilitary force in the country. It patrols India-Myanmar border and insurgency in North-East.
Border Security Force (BSF)	It was established in 1965. It patrols Pakistan and Bangladesh.

National Cadet Corps (NCC)	It was established in 1948. Its main objective is to stimulate interest among the youth in the defence of the country in order to build up a reserve manpower to expand armed forces.
Central Reserve Police Force (CRPF)	It was set-up in 1939. Its main objective is to assist the State/Union Territory Police in maintenance of law and order. The 88th Battalion of CRPF, known as 'Mahila Battalion' (commissioned on 30th March, 1986) is the world's first paramilitary force comprising entirely of women.
Home Guard	It was established in 1946, to assist the police in maintaining security, to help defence forces and to help local authorities.
Coast Guard	It was set-up in 1978. Its objective is to protect the maritime and other national interests in the maritime zones of India.
Intelligence Bureau (IB)	It was set-up in 1920. Its objective is to collect secret information relating to country's security. It was originally set-up as Central Special Branch (CSB) in 1887 and renamed IB in 1920.
Central Bureau of Investigation (CBI)	It was established in 1953. Its objective is to investigate cases of misconduct by public servants, cases of cheating, embezzlement, fraud also work on International crime cases in collaboration with INTERPOL.
National Crime Records Bureau (NCRB)	It was established in 1986. Its objective is to collect crime statistics at the national level.
Rapid Action Force (RAF)	It was established in 1991, wing of CRPF. It deals with riot, crowd control, relief operation.

DRDO

It is responsible for the development of technology for use by the military and is headquartered in New Delhi, India. Defence Research and Development Organisation (DRDO) was formed in 1958, by the merger of Technical Development Establishment and the Directorate of Technical Development and Production with the Defence Science Organisation.

Integrated Guided Missile Development Programme (IGMDP)

The IGMDP was launched in 1983, for the development of a comprehensive range of missiles including the intermediate range Agni Missile (surface to surface) and short range missiles such as the Prithvi Ballistic Missile (surface to surface), Akash Missile (surface to air), Astra Missile (air to air), Trishul Missile (surface to air) and Nag Missile (anti tank).

Indian Missiles

Agni (I-VI)	Agni series of missile is a ballistic missile having 6 missiles in it with a range varying from 700-10000 km. Agni I-V are in operational stage while VI is in testing and developmental stage.
Prithvi	Prithvi is a tactical surface to surface Short-Range Ballistic Missile (SRBM). It has three versions - Prithvi I, II, III with range of 150-350 km.
Dhanush (SS-350)	Dhanush is reportedly a naval version of Prithvi-III, which can be launched from ships. It can carry 500 kg of conventional or nuclear warhead.
Astra	Astra is Beyond Visual Range (BVR) air to air missile. Astra is designed to be capable of engaging targets at varying range and altitudes allowing for both short-range targets (upto 20 km) and long range targets (upto 80 km) using alternative propulsion modes.
Akash	Akash is a medium range surface to air missile. The missile can target aircraft upto 30 km away, at altitudes of 18000 m. Akash can be fired from both tracked and wheeled platforms. Akash is said to be capable of both conventional and nuclear warheads, with a reported payload of 60 kg.

Trishul	Trishul is a short range surface to air missile. The range of the missile is 12 km and is fitted with a 15 kg warhead. The weight of the missile is 130 kg.
Nag	Nag is India's third generation 'Fire-and-Forget' anti-tank missile. It is an all weather, top attack missile with a range of 3 to 7 km.
Sagarika	K-15 Sagarika is a nuclear-capable submarine launched ballistic missile with a range of 700 km. It belongs to the K Missile family. The latest test of the K-15 Missile was done on 11th March, 2012.
Brahmos	Brahmos is a stealth supersonic cruise missile that can be launched from submarines, ships, aircraft or land. It is a jointly developed by India and Russia. It is the world's fastest cruise missile in operation. The missile travels at a speed of mach 2.8 to 3.0. It has a range of 290 km. It can carry 300 kg of conventional or nuclear warhead.
Shaurya	The Shaurya Missile is a short-range surface to surface ballistic missile developed by DRDO for use by the Indian Army capable of hypersonic speeds, it has a range of 600 km and is capable of carrying a payload of one to one conventional or nuclear warhead. Shaurya Missile is a land version of the under-water launched K-15 Missile, Sagarika.
Nirbhay	Nirbhay is a long range, subsonic cruise missile developed in India. The missile has a range of 1000 km. The missile will have a speed of 0.8 mach. The Nirbhay will be able to be launched from multiple platforms on land, sea and air.
NOVATOR K-100	It is an air to air missile designed as an AWACS killer for ranges upto 300-400 km.
BARAK-8	It is an India-Israel long range surface to air missile.
Ashwin	It is an anti-ballistic missile designed to intercept incoming ballistic missiles in endo-atmosphere at an altitude of 30 km.

Defence Production Undertakings

Name and Established in	Factories at
Bharat Dynamics Ltd (BDL) 1970	Hyderabad
Bharat Earth Movers Ltd (BEML) 1964	Bengaluru, Mysore, Kolar
Bharat Electronics Ltd (BEL) 1954	Bengaluru, Ghaziabad, Pune, Machilipatnam, Talaja (Maharashtra), Panchkula (Haryana), Koldwara (Uttarakhand), Hyderabad, Chennai
Garden Reach Shipbuilders and Engineers Ltd (GRSE) 1934 (Government acquired in 1960)	Kolkata
Goa Shipyards Ltd (GSL) 1957	Goa
Heavy Vehicles Factory	Avadi
Hindustan Aeronautics Ltd (HAL) 1964	Bengaluru (five factories), Koraput, Nashik, Karwa, Kanpur, Lucknow, Barrackpore, Hyderabad
Mazagaon Docks Ltd (MDL) 1960	Mumbai
Mishra Dhatu Nigam Ltd (MIDHANI) 1973	Hyderabad
Praga Tools	Hyderabad

List of Submarines of the Indian Navy

INS Vibhuti	It was India's first indigenously built missile boat launched at Mazgaon Docks in Mumbai. It is a veer class corvette.
INS Shakti	It was India's first indigenously built submarine. The ship is one of the largest in the Indian Navy as it is 175 m in length and 32 m in width.
INS Kadamba	It is India's largest naval base located near Karwar in Karnataka. It was commissioned in 2005, under the project Seabird.
INS Talwar	It is the leadership of the Talwar class frigates of the Indian Navy. Its name means 'Sword' in Hindi. It was built in Russia and commissioned into the Indian Navy in 2003.
INS Vikrant	The first aircraft carrier of Indian Navy was INS Vikrant. India purchased the INS Vikrant from the United Kingdom in 1957.
INS Mysore	It is a Delhi class guided missile destroyer currently in active service with the Indian Navy. INS Mysore was built at Mazagaon Dock Limited in Mumbai and it was commissioned in 1999.
INS Beas	It is a Brahmaputra class frigate of the Indian Navy. It was commissioned in 2005.
INS Chakra II	The INS Chakra II (K-152 Nerpa) is a 8140 tonne project 518 (NATO Akula II) type nuclear-powered attack submarine.
INS Arihant	Powered by 83 MW PWR. It is a part of Indian Navy's Advanced Technology Vessel.
INS Kalvari	It is a diesel-electric attack Submarine which is designed by DCNS and being manufactured at Mazagaon Dock Limited, Mumbai. It is commissioned in late 2017 (expected) but launched in April 2015. It is Navy's six kalvari-class submarines.

UNITED NATIONS

The United Nations (UN) is an international organisation, whose stated aims are facilitating co-operation in international law, international security, economic development, social progress, human rights and achievement of world peace. The United Nations Day is celebrated on 24th October each year. Presently, there are 193 member states of the United Nations.

Principle Organs

There are six principle organs of the United Nations, which are as follows

1. General Assembly

- Headquarters of General Assembly is at New York. General Assembly consists of all member states of the UN. Each member nation can send five delegates, but each nation has only one vote.
- The General Assembly meets at least once in a year and the session commences on the first Tuesday of September. It appoints the Secretary General of UN Secretariat on the recommendation of the Security Council.
- The presidency of the Assembly rotates each year among the five geographical groups of the countries viz Asia, African, Latin America, East European and West European and other states.

2. Security Council

- The main aim of Security Council is the maintenance of the international peace and security. The Security Council originally consisted of eleven members, but increases to fifteen in 1965.
- Security Council comprises of five permanent members namely China, UK, Russia, France and USA and 10 non-permanent members, elected for a term of 2 years by a two-third majority of the General Assembly, five non-permanent members retire every year. Retiring members cannot be re-elected immediately.
- Permanent member have veto power, which can be cast against any decision supported by the majority members. The headquarters of Security Council is at New York (US).

3. Economic and Social Council (ECOSOC)

Its main aim is to promote social progress and better standards of life. ECOSOC comprises 54 members, 18 (one third) of whom are elected every year by the General Assembly of UN, to serve a 3 years term. The retiring members as well as the President are eligible for immediate re-elections. The headquarters of ECOSOC is at New York (US).

4. The Trusteeship Council

- The United Nations Trusteeship Council was established to help ensure that trust territories were administered in the best interests of their inhabitants and of international peace and security.
- Trusteeship Council was formed in 1945. The headquarters of Trusteeship Council was at New York (US). The Trusteeship Council suspended operation on 1st November, 1994. With the independence of Polau, the last remaining United Nations trust territory, on 1st October, 1994.

5. The International Court of Justice (ICJ)

- The International Court of Justice (ICJ) is the primary judicial organ of the United Nations. The ICJ was established in 1945. It consists of 15 judges. The judges of the court are elected by the General Assembly along with the Security Council for a 9 years term.
- Its main functions are to settle legal disputes submitted to it by states and to provide advisory opinions on legal questions submitted to it by only authorised international organs, agencies and the UN General Assembly. The headquarters of ICJ is at Hague (Netherlands).

6. The Secretariat

- The United Nations Secretariat is one of the principle organs of the United Nations, an inter governmental organisation charged with aiding/helping states to collectively maintain international peace and security. It serves as a forum for member-states to discuss and resolve pressing issues in the international field through primarily diplomatic resources.
- The Secretariat is composed of a Secretary General, assisted by a staff of international civil servants worldwide. The Secretary General is appointed by the General Assembly upon the recommendation of the Security Council.
- The Secretary General of the UN is elected for 5 years and eligible for re-election, although, none so far has held office for more than two terms.
- The present Secretary General of the UN is Antonio Guterres.

Important International Agencies

Name of Agency	Estd in	Headquarter	Objectives
International Telecommunication Union (<i>ITU</i>)	1865	Geneva	Sets international regulations for radio telegraph, telephone and space radio communications.
International Labour Organisation (<i>ILO</i>)	1919	Geneva	To improve conditions and living standard of workers.
International Monetary Fund (<i>IMF</i>)	1945	Washington DC	Promotes international monetary co-operation.
Food and Agricultural Organisation (<i>FAO</i>)	1945	Rome	To improve living conditions of rural population.
International Bank for Reconstruction and Development (<i>IBRD</i>)	1944	Washington DC	To provide funds from different sources, offers loans to middle income developing countries.
United Nations Educational, Scientific and Cultural Organisation (<i>UNESCO</i>)	1945	Paris	To promote collaboration among nations through education, science and culture.
United Nations International Children's Emergency Fund (<i>UNICEF</i>)	1946	New York	To promote children's welfare all over the world.
World Health Organisation (<i>WHO</i>)	1948	Geneva	Attainment of highest possible level of health by all people.
International Atomic Energy Agency (<i>IAEA</i>)	1957	Vienna	To promote peaceful uses of atomic energy.
International Development Association (<i>IDA</i>)	1960	Washington DC	An affiliate of the World Bank, aims to help under-developed countries raise living standards.
United Nations Environmental Programme (<i>UNEP</i>)	1972	Nairobi	Promotes international co-operation in human environment.
World Trade Organisation (<i>WTO</i>)	1995	Geneva	Setting rules for world trade to reduce tariffs.

International Organisations and Groups

Organisations and HQ	Members	Objectives
Red Cross Estd in 1863 (Geneva, Switzerland)	190	International Humanitarian Movement for relief of suffering in time of war/disaster.
International Criminal Police Organisation (INTERPOL) Estd in 1923 (Lyon, France)	190	To promote international co-operation among criminal police authorities.
Arab League (AL) Estd in 1945 [Cairo (Egypt)] Syria suspended following the 2011 uprising	22	To promote economic, social, political and military co-operation.
International Organisation for Standardisation (ISO) Estd in 1947 (Switzerland)	163	To promote the development of international standards.
North Atlantic Treaty Organisation (NATO) Estd in 1949 (Brussels)	28	Mutual defence and co-operation
South-East Asia Treaty Organisation (SEATO) Estd in 1954 (Bangkok)	8	To provide for collective and economic co-operation in South-East Asia.
Organisation of Petroleum Exporting Countries (OPEC) Estd in 1960 [Vienna (Austria)]	14	Attempts to set world prices by controlling oil production and also pursues member interest in trade and development.
World Wildlife Fund For Nature (WWF) Estd in 1961 (Switzerland)	–	To save the wildlife from extinction.
Amnesty International (AI) Estd in 1961 (London)	–	To keep a watch over human rights violation worldwide. Got Nobel Prize in 1977 for peace.
Non-Aligned Movement (NAM) Estd in 1961	114	Political co-operation and establishment of separate identity from both USA and USSR (in the Cold-War era).
Association of South-East Asian Nations (ASEAN) Estd in 1967 (Jakarta)	10	Regional economic social and cultural co-operation among the non-communist countries of South-East Asia.
Group of 8 (G-8) on 24th March, 2014, Russia was suspended, due to association with crimean crises	8	To promote co-operation among major non-communist economic power.
World Economic Forum (WEF) Estd in 1971 (Geneva)	–	To improve the state of the world by engaging leaders in partnerships to shape global, regional and industry agendas.
Gulf Co-operation Council (GCC) Estd in 1981	6	It is a political and economic union of the Arab states.
South Asian Association for Regional Co-operation (SAARC) Estd in 1985 (Kathmandu)	8	To promote economic, social and cultural co-operation.
Group of 15 (G-15) Estd in 1989	17	To promote economic co-operation among developing nations.
Asia Pacific Economic Co-operation (APEC) Estd in 1989 (Singapore)	21	To promote trade and investment in the Pacific basin.
Group of Twenty (G-20) Estd in 1999-2014 summit-Brisbane	20	For co-operation and consultation on matters pertaining to the International Financial System.
BRICS (Brazil, Russia, India, China and South Africa) First formal summit-Yekaterinburg, 2009	5	To improve the economical condition of the country.